Interactions and Cooperation
(Chapter 9)

Colm Connaughton
London Mathematical Laboratory

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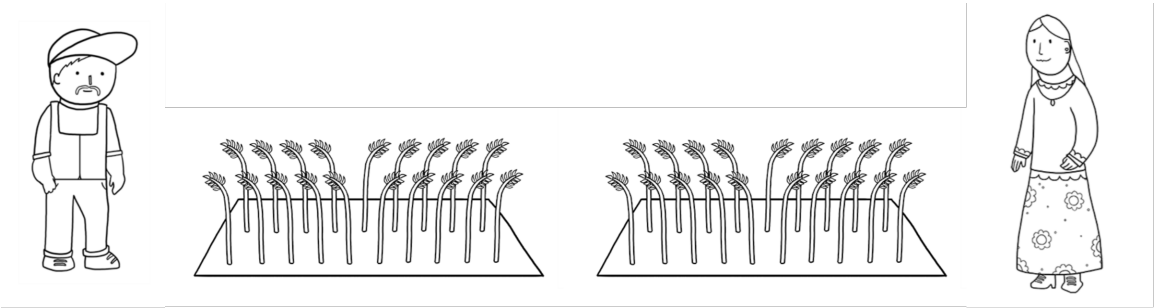
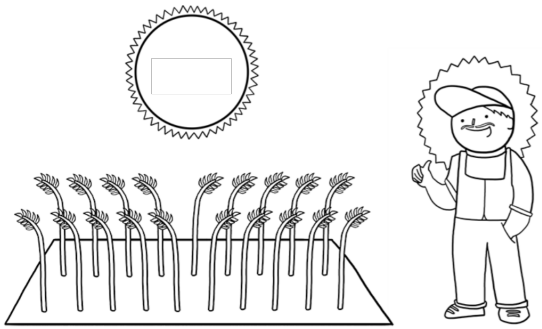
The Farmers Fable

See: `https://www.farmersfable.org/`
Ann and Bill are farmers.



Good season:



Bad season:

$$X_{t+1} = (g + r)\, X_t$$

$$X_{t+1} = (g - r)\, X_t$$

The Farmers Fable

Ann and Bill have good and bad seasons independently with probability $\frac{1}{2}$.
Working independently, their wealths each follow the coin toss dynamic:

$$X_{t+1} = \begin{cases} (g+r)\,X_t & \text{with probabilty } \frac{1}{2} \\ (g-r)\,X_t & \text{with probabilty } \frac{1}{2}. \end{cases}$$

Expected growth factor is $\mathbb{E}\left[G\right]_{\text{indiv}} = g$:

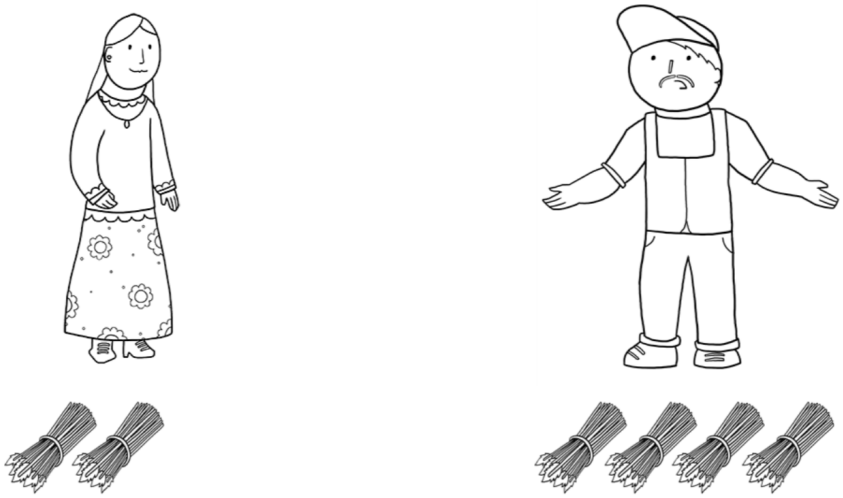
$$\mathbb{E}\left[W_t\right] = W_0\, g^t$$

Time averaged growth factor is $\bar{G}_{\text{indiv}} = \sqrt{g^2 - r^2}$:

$$\text{ML}\left[W_t\right] = W_0 \left(\sqrt{g^2 - r^2}\right)^t.$$

Pooling and sharing

Now imagine that Ann and Bill make a pact. At the end of every season they pool their harvests and share them evenly.



Outcome	Growth factor	Probability
Both have bad seasons	$g - r$	$\frac{1}{4}$
One good, one bad	g	$\frac{1}{2}$
Both have good seasons	$g + r$	$\frac{1}{4}$

Expected value vs EE analysis

The expected growth factor per season is:

$$\mathbb{E}\left[G\right]_{\text{coop}} = \frac{1}{4}\left(g - r\right) + \frac{1}{2}g + \frac{1}{4}\left(g + r\right) = g = \mathbb{E}\left[G\right]_{\text{indiv}}.$$

There is no benefit to pooling and sharing?

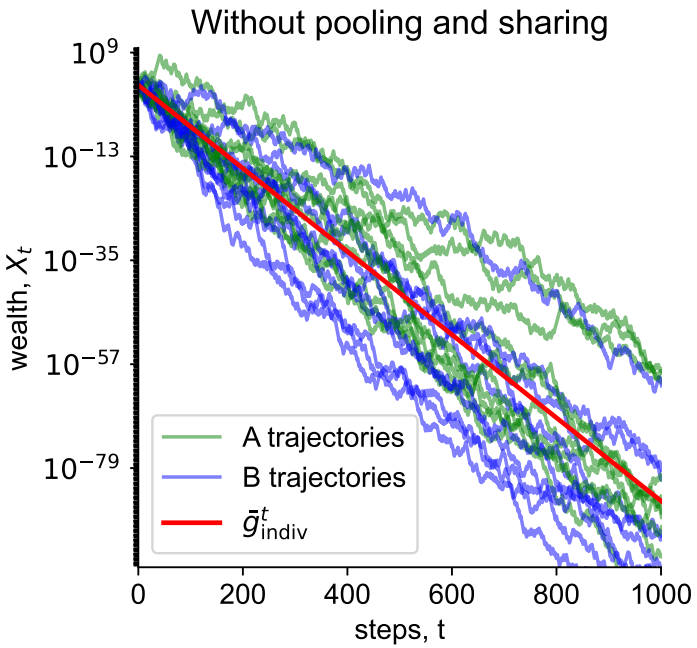
The time-averaged growth factor is:

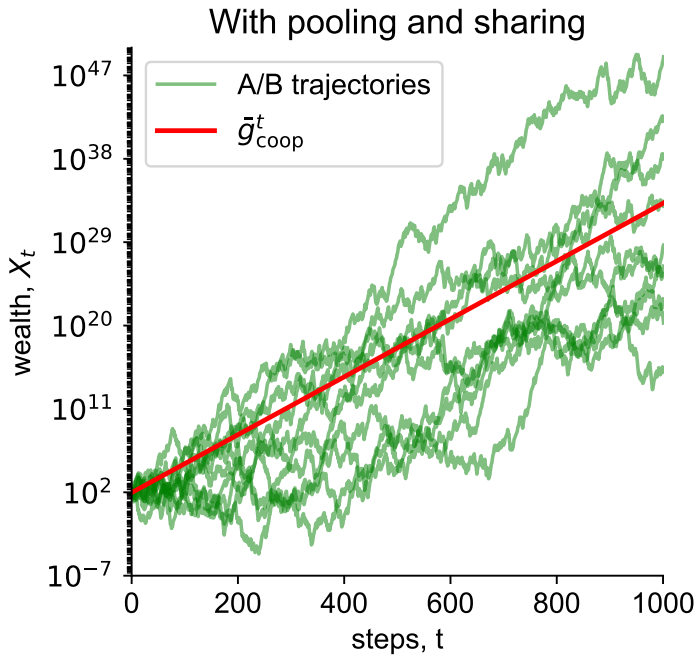
$$\bar{G}_{\text{coop}} = \sqrt{g\sqrt{g^2 - r^2}}$$

Can show that $\bar{G}_{\text{coop}} > \bar{G}_{\text{indiv}}$ for all allowed values of g and r. Pooling and sharing increases the growth rate.

An example where survival requires cooperation

Choose $g = \sqrt{2}$, $r = \frac{2}{\sqrt{3}}$.
$G_{\text{indiv}} < 1$ but $G_{\text{coop}} > 1$





Generalisation to N members and continuum limit

Wealths $X_1(t)$, $X_2(t), \ldots X_N(t)$.
Introduce random growth factors

$$g_i(t) = \mu\,\Delta t + \sigma \Delta W_i(t)$$

where $\Delta W_i(t) \sim \mathcal{N}(0, \Delta t)$, independent.
Each round with full pooling and sharing, each agent gets

$$\begin{aligned}
\frac{1}{N}\sum_{i=1}^{N} g_i(t)\,X_i(t) &= \frac{1}{N}\sum_{i=1}^{N} \left(\mu\,\Delta t + \sigma \Delta W_i(t)\right)\,X_i(t) \\
&= \left(\mu \Delta t + \frac{\sigma}{N}\sum_{i=1}^{N} \Delta W_i(t)\right)\,X_i(t) \\
&= \left(\mu \Delta t + \frac{\sigma}{\sqrt{N}} \Delta W(t)\right)\,X_i(t)
\end{aligned}$$

where $\Delta W(t) \sim \mathcal{N}(0, \Delta t)$.

Generalisation to N members and continuum limit

Single representative agent:

$$X(t+\Delta t) = X(t) + X(t)\left(\mu\Delta t + \frac{\sigma}{\sqrt{N}}\Delta W(t)\right)$$

Continuum limit ($\Delta t \to 0$):

$$dX(t) = X(t)\left(\mu\, dt + \frac{\sigma}{\sqrt{N}}\, dW_t\right)$$

Time averaged growth rate is

$$\bar{g}^{(N)} = \mu - \frac{\sigma^2}{2N}.$$

Larger cooperatives grow faster.

Heterogeneity

Suppose agent has it's own individual growth rate:

$$\bar{g}_i = \mu_i - \frac{\sigma_i^2}{2}$$

The growth rate of the cooperative is

$$\bar{g}^{(N)} = \frac{1}{N} \sum_{i=1}^{N} \left(\mu_i - \frac{\sigma_i^2}{2N} \right)$$

Cooperation is beneficial to individuals if $\bar{g}^{(N)} > \bar{g}_i$ for $i = 1 \ldots N$. This is not guaranteed.
For example, if $N = 2$, cooperation is beneficial if

$$|\bar{g}_1 - \bar{g}_2| < \frac{\sigma_1^2 + \sigma_2^2}{4}.$$

Correlated fluctuations

Also consider what happens when fluctuations $\Delta W_i(t)$ are correlated via correlation matrix

$$\rho_{ij} = \begin{cases} 1 & \text{if } i = j \\ \rho & \text{if } i \neq j. \end{cases}$$

The growth rate of the cooperative is shown to be

$$\bar{g}^{(N)}(\rho) = \mu - \frac{\sigma^2}{2N}\left(1 + (N-1)\rho\right).$$

Cooperation benefit disappears as $\rho \to 1$.

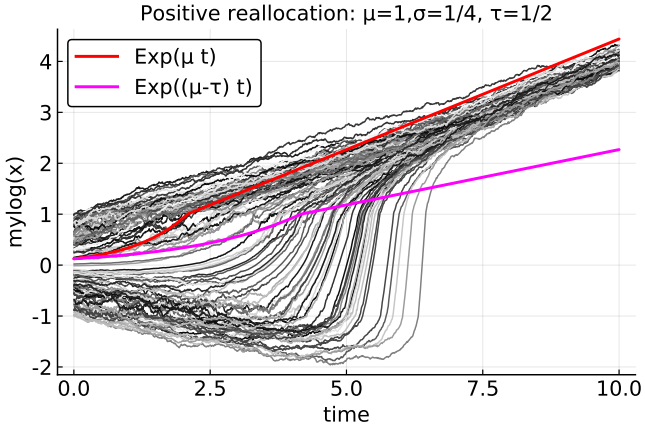
Reallocation: partial pooling and sharing

Reallocating Geometric Brownian Motion:

$$dX_i = X_i(t)\left((\mu - \tau)\,dt + \sigma\,dW(t)\right) + \tau\langle X(t)\rangle_N$$

where $\langle X(t)\rangle_N$ denotes the finite ensemble average:

$$\langle X(t)\rangle_N = \frac{1}{N}\sum_{i=1}^{N} X_i(t).$$



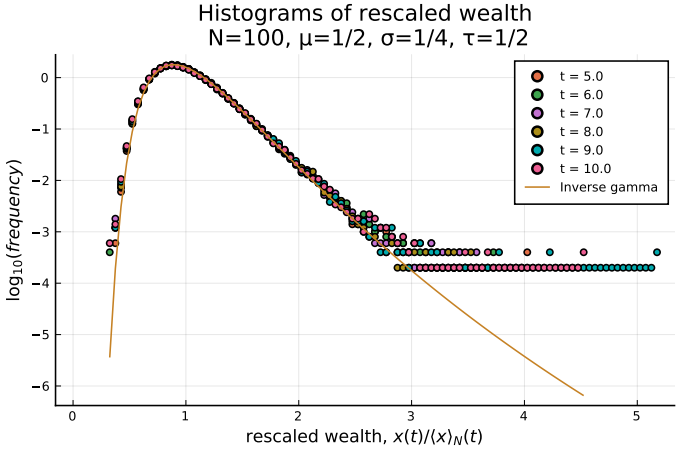
- All agents contribute to shared pot at rate τ and receive an equal share of the pot back.
- Wealth $x_i(t)$ can become negative.
- Can show that

$$\langle X(t)\rangle_N = X(0)\,e^{\mu\,t}.$$

Positive reallocation

Define rescaled wealth:

$$w_i(t) = \frac{X_i(t)}{X(0)\, e^{\mu\, t}}$$



- If $\tau > 0$ can show that $w_i(t)$ becomes stationary for large times
- Distribution of rescaled wealth is inverse gamma distributed:

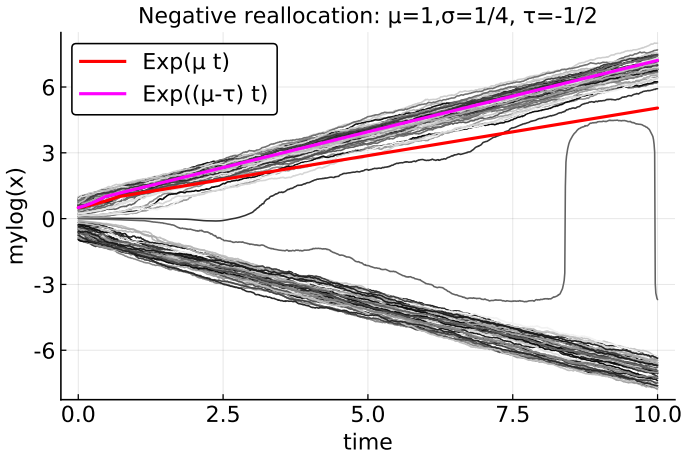
$$P(w) = \frac{(\zeta - 1)^{\zeta}}{\Gamma(\zeta)} e^{-\frac{\zeta - 1}{w}}\, w^{-(\zeta+1)}$$

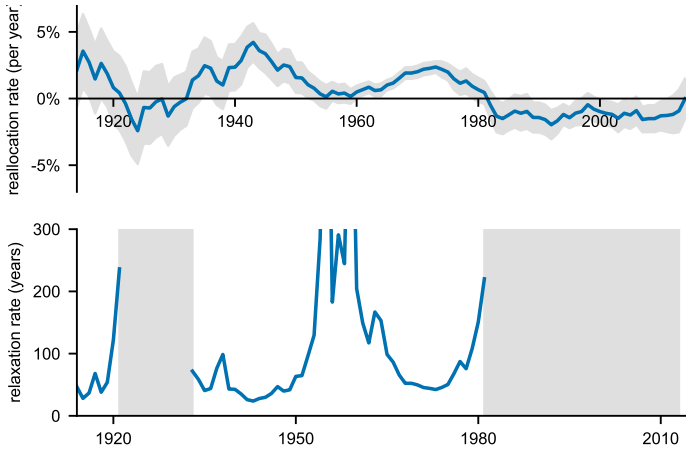
where $\zeta = 1 + \frac{2\tau}{\sigma^2}$.

- Note that redistribution creates a power law tail.

Negative reallocation

Empirical estimation of the τ parameter from US wealth data suggests that τ can be negative.





Mathematically little is known about the negative τ regime other than simulations.

Take home messages

- Cooperation in not surprising: pooling and sharing among homogeneous agents doesn't just reduce risk, it increases growth rates for all.
- This does not imply a communist utopia: in a population of heterogeneous agents there are restrictions on cooperation benefits.
- Mechanism specifically relies on volatility reduction over time: no claim that this in the only mechanism for cooperation among self-interested individuals.
- Redistribution (partial pooling and sharing) creates power law tails in the relative wealth distribution. There is a mismatch with the conventional interpretation of Pareto vs lognormal tails as measures of inequality.
- Negative redistribution is an empirically interesting regime and is poorly understood quantitatively.